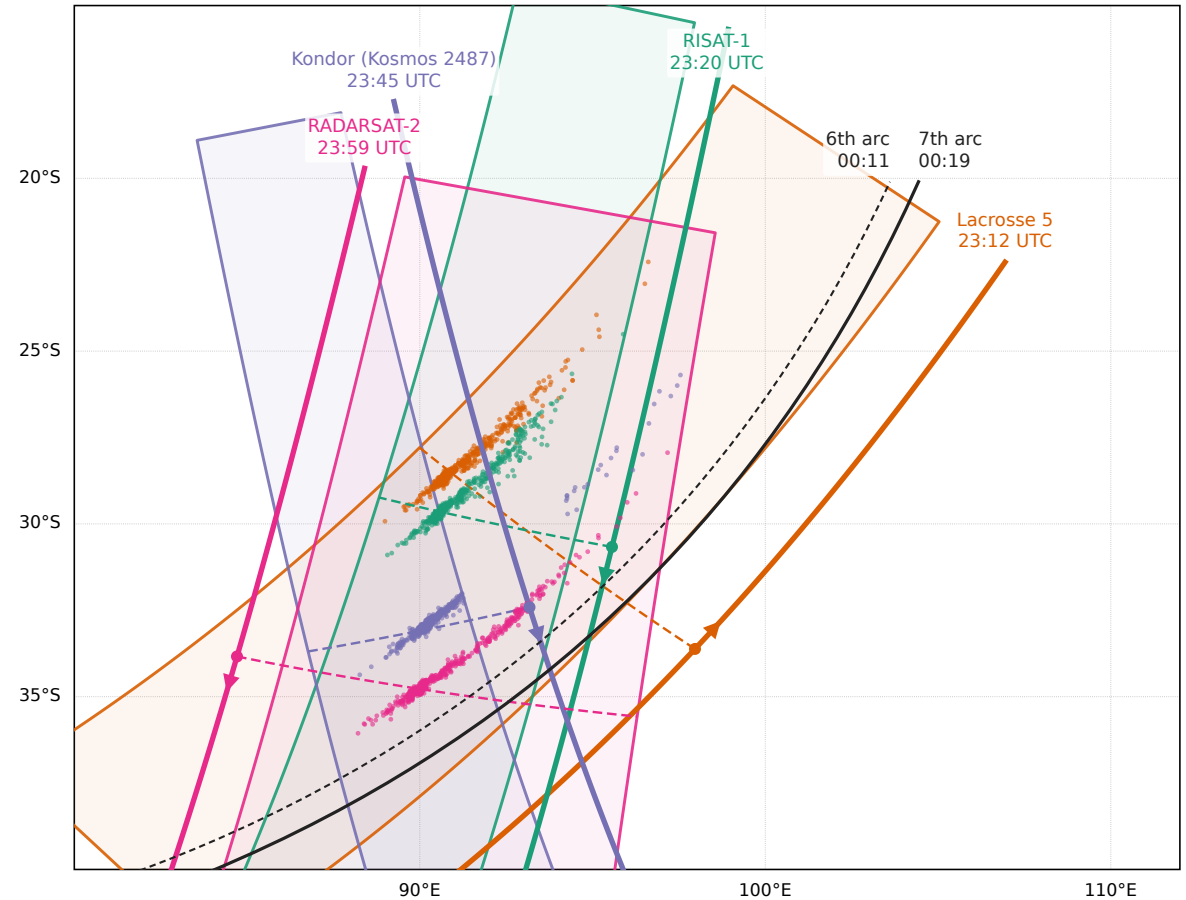
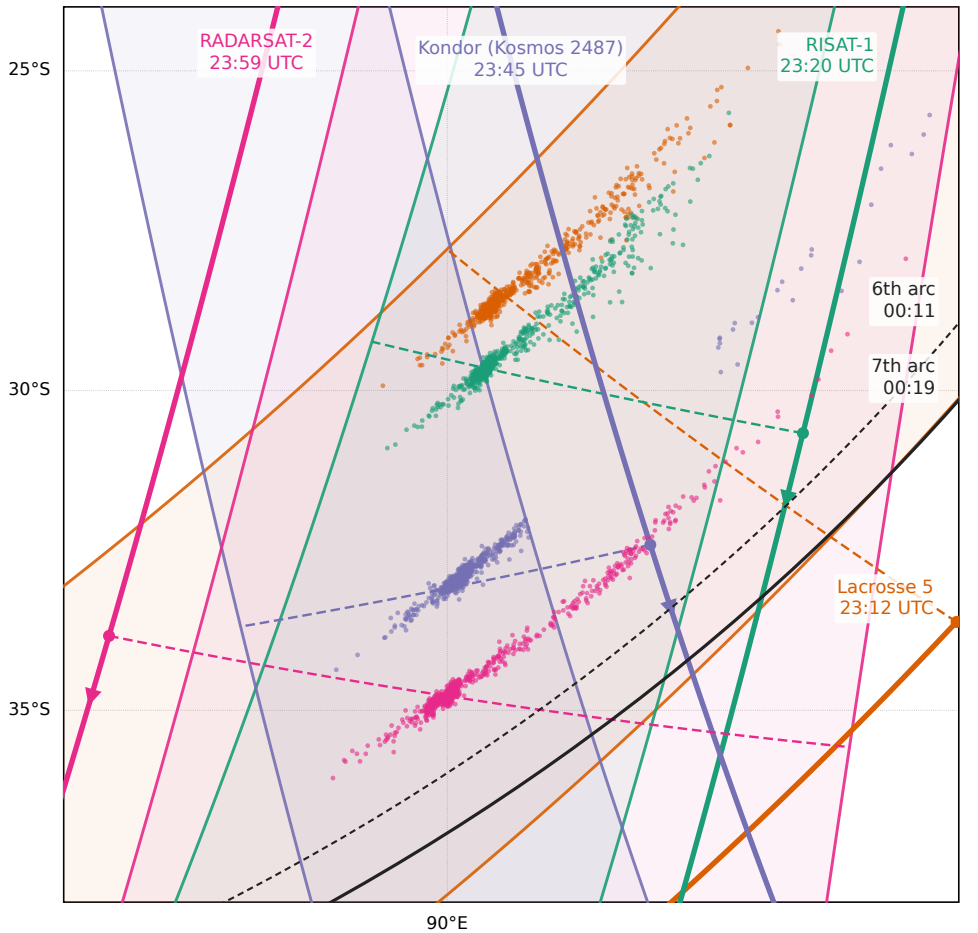
Where four radar satellites could have imaged MH370 in flight, 7–8 March 2014

Thick coloured line with arrow: path of the point below the satellite, 4 min either side of the moment it swept over the aircraft. Outlined area of the same colour: the radar's reach. These radars look to one side only, so the reach lies wholly to one side of the line, 100–1,100 km from it. Dashed line with dot: the line straight across the track at the moment of the sweep, out to the far edge of the reach. Dots: posterior aircraft positions at that moment. Black solid: 7th arc; black dashed: 6th arc.

a Posterior aircraft positions when each radar swept over them, with the radar's reach



b Detail of the southern track (dots: route samples inside each radar's reach)



Plain-language note: each colour is one radar satellite. The dots show where the aircraft probably was when that satellite passed; they lie inside its outlined reach, so it could have imaged the aircraft if it had been ordered to. Lacrosse 5 (USA) at 23:12, RISAT-1 (India) at 23:20, Kondor, Kosmos 2487 (Russia) at 23:45, RADARSAT-2 (Canada) at 23:59 UTC. Nothing here shows that an image was taken.

Technical: core run J routes.npy, waypoints family (48,000 routes, P(family) 0.9982), linear interpolation along 10-min chords (about 25 NM tolerance at turns); 800 in-reach samples drawn per pass. SGP4 on March 2014 element sets, 5 s steps; reach = ground offsets at the minimum and maximum incidence on the side the aircraft lay (RADARSAT-2 left, 10–60°, 125–1,078 km; RISAT-1 right, 12–55°, 107–670 km, grade B; Lacrosse 5 left, 20–60°, 236–999 km; Kosmos 2487 right, 20–55°, 166–615 km, grade C). Cross-track line drawn at the median sweep time of the in-reach routes; times rounded to the minute. Probability inside reach: 1.00, 1.00, 0.80, 0.99. FL350 assumed; the radar would place the aircraft 7–18 km toward itself and 2–18 km along its track from the true position.